

Nmap Complete Guide

Beginner to Advanced — explained in English

Educational and legal-use notice

This guide is for educational and learning purposes. Use Nmap only on your own devices/networks or where you have explicit legal permission. Scanning another network without permission may be illegal.

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1. What is Nmap and why is it used?

Nmap (Network Mapper) is a free and open-source tool used for network scanning. Its main job is to gather information about a device such as a computer, router, phone, or server: whether it is online, which ports are open, which services are running, and sometimes which operating system is in use.

Think of a building as an IP address and its many doors as ports. Nmap checks those doors to see whether they are open or closed and what service is behind an open door.

Common uses

- Network administrators monitor their own networks.
- Security professionals use it for authorized testing and vulnerability discovery.
- Students learn cybersecurity in their own practice labs.
- System administrators check whether unwanted services are running.

2. Installing Nmap

Kali Linux commonly includes Nmap. To check or install/update it:

```
nmap --version
```

```
sudo apt update
```

```
sudo apt install nmap -y
```

The first command shows whether Nmap is installed and its version. The update command refreshes package information; the install command installs Nmap.

3. Basic Scanning Commands

3.1 Simple scan

```
nmap <target>
```

Replace <target> with an IP address or domain name. By default, this checks the most common 1000 ports.

3.2 Scan a complete subnet/network

```
nmap <network>/24
```

A /24 can cover up to 256 IP addresses. Only use this on a network you own or are authorized to test.

3.3 Scan multiple targets

```
nmap 10.0.0.1 10.0.0.2 10.0.0.3
```

```
nmap 10.0.0.1-50
```

The first command scans multiple specified IPs; the second scans an address range.

4. Types of Port Scans

4.1 TCP Connect Scan

```
nmap -sT <target>
```

Nmap completes a normal TCP connection. It usually does not require root/admin privileges, but it can be slower and more visible in logs.

4.2 SYN Scan (Stealth Scan)

```
sudo nmap -sS <target>
```

Nmap starts the TCP connection but does not complete it. It is generally faster and can leave less connection-level trace, and normally requires root/sudo privileges.

4.3 UDP Scan

```
sudo nmap -sU <target>
```

Checks UDP ports, used by services such as DNS and DHCP. UDP scanning is generally slower than TCP scanning.

4.4 Scan specific ports

```
nmap -p 22,80,443 <target>
```

```
nmap -p 1-100 <target>
```

```
nmap -p- <target>
```

The first scans selected ports, the second scans ports 1–100, and -p- scans all 65,535 TCP ports.

4.5 Fast scan

```
nmap -F <target>
```

Checks the top 100 common ports for a quicker result.

5. Host Discovery Commands

5.1 Check whether devices are online

```
nmap -sn <target/24>
```

This performs host discovery without a normal port scan and can quickly identify live devices.

5.2 Skip ping discovery

```
nmap -Pn <target>
```

If a firewall blocks ping, Nmap may think a device is down. -Pn skips host-discovery checks and proceeds with scanning.

6. Scan Timing and Speed

Nmap has six timing templates:

- T0 Paranoid — very slow
- T1 Sneaky — slow and less detectable
- T2 Polite — uses less network bandwidth
- T3 Normal — default speed
- T4 Aggressive — faster on stable networks
- T5 Insane — fastest, but results can sometimes be missed

Example:

```
nmap -T4 <target>
```

7. OS and Version Detection

7.1 Service version detection

```
nmap -sV <target>
```

Attempts to identify the software and version running on open ports.

7.2 Operating-system detection

```
sudo nmap -O <target>
```

Attempts to estimate the target operating system from network responses. It is not 100% accurate.

7.3 Aggressive scan

```
sudo nmap -A <target>
```

Enables OS detection, version detection, script scanning, and traceroute together. It provides more detail and may take longer.

8. Nmap Scripting Engine (NSE)

Nmap includes many scripts that can automatically check for vulnerabilities, misconfigurations, and additional information.

8.1 Default scripts

```
nmap -sC <target>
```

Runs the default script category.

8.2 Specific script

```
nmap --script=<script-name> <target>
```

Example:

```
nmap --script=http-title <target>
```

This can retrieve a web page title.

8.3 Vulnerability scripts

```
nmap --script vuln <target>
```

Runs scripts in the vuln category to check for known vulnerabilities. Use only against authorized targets.

9. Saving Output Formats

```
nmap -oN result.txt <target>
```

Normal readable text output.

```
nmap -oX result.xml <target>
```

XML output for use with other tools.

```
nmap -oG result.gnmap <target>
```

Grepable output.

```
nmap -oA result <target>
```

Saves Normal, XML, and Grepable formats using the same prefix.

10. Firewall/IDS Evasion Options

This section is conceptual and intended for authorized penetration testing only.

```
nmap -f <target>
```

Fragments packets; behavior depends on the network and security device.

```
nmap -D RND:5 <target>
```

Uses five random decoy addresses along with the scan.

```
nmap --data-length 25 <target>
```

Adds extra data to packets. Do not use evasion features against systems without permission.

11. Useful Combinations and Real Examples

Detailed scan:

```
sudo nmap -sS -sV -O -p- -T4 <target>
```

Combines SYN scan, service-version detection, OS detection, all ports, and faster timing.

Find live devices:

```
nmap -sn <network>/24
```

Web-server check:

```
nmap -p 80,443 -sV --script=http-title <target>
```

Quiet/fast scan:

```
nmap -T4 -F <target>
```

12. Quick Reference Cheat Sheet

`nmap <target>` — basic scan

`nmap -sn <target>` — host discovery

`nmap -p- <target>` — all 65,535 ports

`nmap -sV <target>` — service-version detection

`nmap -O <target>` — OS detection

`nmap -A <target>` — detailed/aggressive scan

`nmap -T4 <target>` — faster timing

`nmap -Pn <target>` — skip ping discovery

`nmap -sC <target>` — default scripts

`nmap -oN file.txt <target>` — save results

13. Ethical and Legal Considerations

Nmap is a powerful tool, so it should be used responsibly.

- Scan only your own devices/networks.
- Get written permission before testing another network.
- In many countries, unauthorized network scanning can violate cyber laws even if no damage occurs.
- For safe practice, use authorized learning labs such as TryHackMe and Hack The Box.
- This guide is educational and is intended to help readers understand Nmap commands and concepts.